Mohamed ElBialy

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PROFESSIONAL EXPERIENCE

AI Engineer · M.I.A Robotics *September 2025 – Present | Alexandria, EG · Hybrid*

- **VLM scene understanding**: developed perception pipeline for robot environment awareness using vision-language models.
- **Re-identification**: applied clustering & model analysis for AnimalCLEF wildlife re-ID competition.

AI Engineer Intern · Electro Pi *May 2026 – August 2026 | Cairo, EG · Hybrid*

- Developing real-time **defect detection** and **segmentation** pipelines using **YOLOv11** to identify manufacturing defects in production environments.
- **Agentic NLP Chatbot:** Developing an AI agent-powered customer support pipeline (**Gemini** / **Hugging Face** LLMs) with session memory, multimodal image retrieval, and automated human-in-the-loop routing.

Computer Vision Engineer Intern · Cellula Technologies *June 2025 – August 2025 | Giza, EG · Remote*

- Developed **3 computer vision applications**, designing and training models from scratch.
- Projects included **image classification** using custom and pretrained CNN models, **satellite image segmentation**, and a **video-based theft detection system**.

EDUCATION

B.Sc. Computer Engineering *September 2022 – June 2027*

Alexandria Higher Institute of Engineering & Technology

Relevant Courses: Linear Algebra, Probability & Statistics, Data Structures, Algorithms, Databases, Operating Systems

TECHNICAL SKILLS

Languages: Python · C++ · SQL · JavaScript

ML / AI: PyTorch · TensorFlow · Scikit-learn · Hugging Face · OpenCV · YOLO · RAG · LLMs · NLP · Prompt Engineering · Fine-Tuning LLMs · LangChain · LangGraph · LangSmith

Data: Pandas · NumPy · Matplotlib · Seaborn · Plotly

Backend & DB: FastAPI · MySQL · PostgreSQL · SQLAlchemy · Alembic

Tools: Docker · Git · Linux · Jupyter

TECHNICAL PROJECTS

SHATO — Voice-Controlled Robotics Assistant ↗ GitHub

- End-to-end **STT** → **LLM** → **TTS** pipeline running **Llama-3.2-3B-Instruct** locally via **llama.cpp** (Q4_K_S quantization) for efficient on-device inference.
- Architected modular microservices with **FastAPI + Docker** Compose and schema-validated safety layer for reliable robot command execution.

UniAssist — Learning Assistant ↗ GitHub

- Developed a bilingual (English/Arabic) AI-powered university assessment and tutoring platform using **Next.js, FastAPI, PostgreSQL, and Docker**, enabling task management, rubric-based evaluation, file submissions, and controlled grade release.
- Implemented **AI tutoring, versioned grading guidance, GitHub-based submission evidence, team collaboration, and role-based workflows** for students, TAs, and professors, with persistent conversations and staff-reviewed AI insights.

AnimalCLEF26 — Wildlife Re-Identification Pipeline ↗ GitHub Publication

- Built end-to-end re-ID pipeline combining **WildFusion** (MiewID global features + ALIKED/DISK local features with LightGlue), species-specific **OpenCV preprocessing**, **k-reciprocal re-ranking**, and **Chinese Whispers** graph clustering.
- **1st place** on AnimalCLEF26 private leaderboard (score: **0.70393**) out of all competing teams at **CVPR & CLEF 2026.**

Adversarial Attacks & XAI on CNNs ↗ Kaggle

- Demonstrated **FGSM adversarial attacks** on CNNs and applied **Grad-CAM (XAI)** to visualize and interpret model vulnerabilities.
- Hardened model via **adversarial training** — reduced Attack Success Rate from **79.5%** → **36.2%** while preserving clean accuracy at **90.5%**.

LEADERSHIP

Head of AI — Microsoft Student Branch AIET

- Led 20+ hands-on AI sessions and organized workshops covering practical machine learning and deep learning topics.
- Mentored branch members, guiding them through practical projects and supporting their professional development.

COMPETITIONS

ECPC 2024

- Ranked **39th** out of 500+ teams in the Egyptian Collegiate Programming Contest — regional ICPC qualifier.